Δ vs frozen (good-oriented, value + 95% CI) · PRETRAIN block / SURGERY block · SCORE = mean per-metric-normalized (0–1) · BLUE BOX + big Δ = best arm per metric
WINNER row (champion duel — S=surgery, P=pretrain, ==tie): surgery 3 · pretrain 5 · tie 0 → PRETRAIN

	action_top1 ↑ better classify motion class from frozen features	motion_cos ↑ better same-motion clips cluster vs different motions	taxonomy_f1 ↑ better read scene attributes from features	future_mse ↓ better predictor guesses masked next-frame tokens	rollout ↓ better error growth over multi-step rollout	causal ↓ better predict 2nd half from 1st half	tdist ↓ better accuracy decay vs prediction horizon	teacher_free ↓ better free-run minus teacher-forced gap	maskratio ↓ better degradation under heavier masking	SCORE (norm 0-1)
ViT-g·2.0 pretrain_2X encoder	+11.62 [+9.260, +13.918]	**+0.221** [+0.215, +0.226]		-0.138 [-0.142, -0.135]	-0.0005 [-0.001, -0.000]	-0.120 [-0.123, -0.116]	-0.0009 [-0.001, -0.000]	-0.0089 [-0.009, -0.009]	-0.018 [-0.019, -0.018]	0.48
ViT-g·2.0 pretrain encoder	+11.67 [+9.479, +13.918]	+0.105 [+0.102, +0.109]		-0.110 [-0.113, -0.107]	**+0.0030** [+0.003, +0.003]	-0.088 [-0.090, -0.085]	**+0.0031** [+0.003, +0.004]	-0.0019 [-0.002, -0.002]	-0.010 [-0.011, -0.010]	0.61
ViT-g·2.0 pretrain head	+0.603 [-1.589, +2.740]	+0.0018 [+0.002, +0.002]		-0.0003 [-0.001, +0.000]	0 [+nan, +nan]	0 [+nan, +nan]	0 [+nan, +nan]	**0** [+nan, +nan]	**0** [+nan, +nan]	0.46
ViT-g pretrain_2X encoder	+9.86 [+7.507, +12.154]	+0.037 [+0.035, +0.038]		+1.01 [+1.004, +1.009]	-0.013 [-0.013, -0.012]	+0.999 [+0.997, +1.002]	-0.0043 [-0.005, -0.004]	-0.016 [-0.016, -0.015]	-0.012 [-0.012, -0.011]	0.47
ViT-g pretrain encoder	+8.05 [+5.699, +10.411]	+0.034 [+0.033, +0.036]		+1.01 [+1.011, +1.016]	-0.013 [-0.013, -0.013]	+1.00 [+1.002, +1.007]	-0.0041 [-0.005, -0.004]	-0.015 [-0.016, -0.015]	-0.013 [-0.014, -0.013]	0.44
ViT-g pretrain head	+4.38 [+2.247, +6.575]	+0.0037 [+0.003, +0.004]		+1.01 [+1.006, +1.011]	-0.015 [-0.015, -0.015]	+1.00 [+0.998, +1.004]	-0.0032 [-0.004, -0.003]	-0.021 [-0.021, -0.021]	-0.013 [-0.014, -0.013]	0.35
ViT-g·2.0 surgery 3stage_DI encoder	+10.90 [+8.658, +13.206]	+0.113 [+0.110, +0.117]		-0.073 [-0.076, -0.070]	-0.0017 [-0.002, -0.001]	-0.065 [-0.068, -0.062]	+0.0010 [+0.001, +0.001]	-0.019 [-0.019, -0.018]	-0.0078 [-0.009, -0.007]	0.46
ViT-g·2.0 surgery 3stage_DI head	**+11.89** [+9.699, +14.192]	+0.112 [+0.108, +0.115]								0.75
ViT-g·2.0 surgery noDI encoder	+11.29 [+8.986, +13.589]	+0.116 [+0.113, +0.120]		-0.070 [-0.072, -0.067]	-0.0008 [-0.001, -0.000]	-0.062 [-0.065, -0.060]	+0.0005 [-0.000, +0.001]	-0.016 [-0.016, -0.015]	-0.0095 [-0.010, -0.009]	0.47
ViT-g·2.0 surgery noDI head	+10.36 [+8.000, +12.767]	+0.119 [+0.115, +0.122]								0.70
ViT-g surgery 3stage_DI encoder	+9.42 [+7.178, +11.616]	+0.036 [+0.035, +0.038]		**+1.02** [+1.014, +1.020]	-0.013 [-0.014, -0.013]	**+1.01** [+1.003, +1.008]	-0.0048 [-0.005, -0.004]	-0.017 [-0.018, -0.017]	-0.014 [-0.015, -0.013]	0.43
ViT-g surgery 3stage_DI head	+8.11 [+5.973, +10.301]	+0.043 [+0.041, +0.044]		+1.01 [+1.011, +1.016]	-0.013 [-0.013, -0.013]	+1.00 [+1.002, +1.007]	-0.0041 [-0.005, -0.004]	-0.015 [-0.016, -0.015]	-0.013 [-0.014, -0.013]	0.45
ViT-g surgery noDI encoder	+8.27 [+6.027, +10.575]	+0.049 [+0.047, +0.051]		+1.01 [+1.007, +1.012]	-0.013 [-0.013, -0.012]	+0.999 [+0.996, +1.001]	-0.0044 [-0.005, -0.004]	-0.014 [-0.014, -0.013]	-0.0099 [-0.011, -0.009]	0.48
ViT-g surgery noDI head	+8.77 [+6.575, +11.014]	+0.055 [+0.053, +0.057]		+1.01 [+1.010, +1.016]	-0.013 [-0.013, -0.013]	+1.00 [+1.002, +1.007]	-0.0041 [-0.005, -0.004]	-0.015 [-0.016, -0.015]	-0.013 [-0.014, -0.013]	0.46
WINNER	S	P	=	S	P	S	P	P	P	